Total antioxidant capacity from diet and risk of myocardial infarction: a prospective cohort of women.

BACKGROUND: There are no previous studies investigating the effect of all dietary antioxidants in relation to myocardial infarction. The total antioxidant capacity of diet takes into account all antioxidants and synergistic effects between them. The aim of this study was to examine how total antioxidant capacity of diet and antioxidant-containing foods were associated with incident myocardial infarction among middle-aged and elderly women. METHODS: In the population-based prospective Swedish Mammography Cohort of 49-83-year-old women, 32,561 were cardiovascular disease-free at baseline. Women completed a food-frequency questionnaire, and dietary total antioxidant capacity was calculated using oxygen radical absorbance capacity values. Information on myocardial infarction was identified from the Swedish Hospital Discharge and the Cause of Death registries. Hazard ratios (HR) and 95% confidence intervals (CI) were calculated using Cox proportional hazard models. RESULTS: During the follow-up (September 1997-December 2007), we identified 1114 incident cases of myocardial infarction (321,434 person-years). In multivariable-adjusted analysis, the HR for women comparing the highest quintile of dietary total antioxidant capacity to the lowest was 0.80 (95% CI, 0.67-0.97; P for trend=0.02). Servings of fruit and vegetables and whole grains were nonsignificantly inversely associated with myocardial infarction. CONCLUSIONS: These data suggest that dietary total antioxidant capacity, based on fruits, vegetables, coffee, and whole grains, is of importance in the prevention of myocardial infarction.